Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number: 2520030237

Name: Hansika Reddy Gunapati

Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

Enabling Multi-Factor Authentication to Secure Your Access

Course	Cloud Infrastructure and Services (25CS2206E)
Experiment	1(b): Creating an IAM user and enabling MFA

1. Aim

To enable Multi-Factor Authentication (MFA) for an AWS IAM user to provide an additional layer of security during the login process.

2. Requirements

Requirement	Details
Web browser	Google Chrome or Microsoft Edge
AWS account	An active AWS account with the root user credentials
IAM user	StudentUser01 (example name) with AdministratorAccess permission
MFA device name	StudentMFA01 (example name)
MFA method	Passkey / Security key, or an Authenticator app (Google Authenticator or Microsoft Authenticator) on a smartphone

3. Theory

Identity and Access Management (IAM) is the AWS service used to control who can sign in to an account and what they are allowed to do. Multi-Factor Authentication adds a second proof of identity to the username and password, so a stolen password alone cannot be used to sign in.

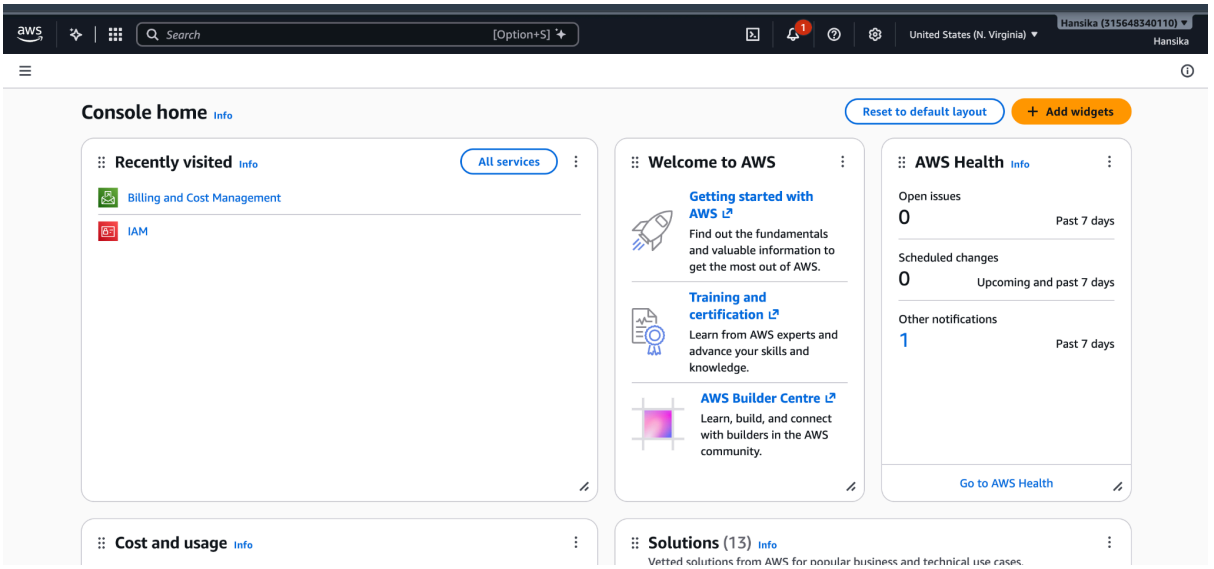
- **Root user:** The owner identity of the AWS account. It should be used only for initial set-up, and day-to-day work is done with IAM users.
- **IAM user and policy:** An IAM user is an identity inside the account. A policy such as AdministratorAccess defines the permissions given to that user.
- **Passkey / Security key:** A phishing-resistant method in which the user verifies with a biometric, device PIN or a physical security key.
- **Authenticator app:** An app on the smartphone that generates a six-digit code which changes every 30 seconds or so.

4. Procedure

Part A: Creating an IAM user

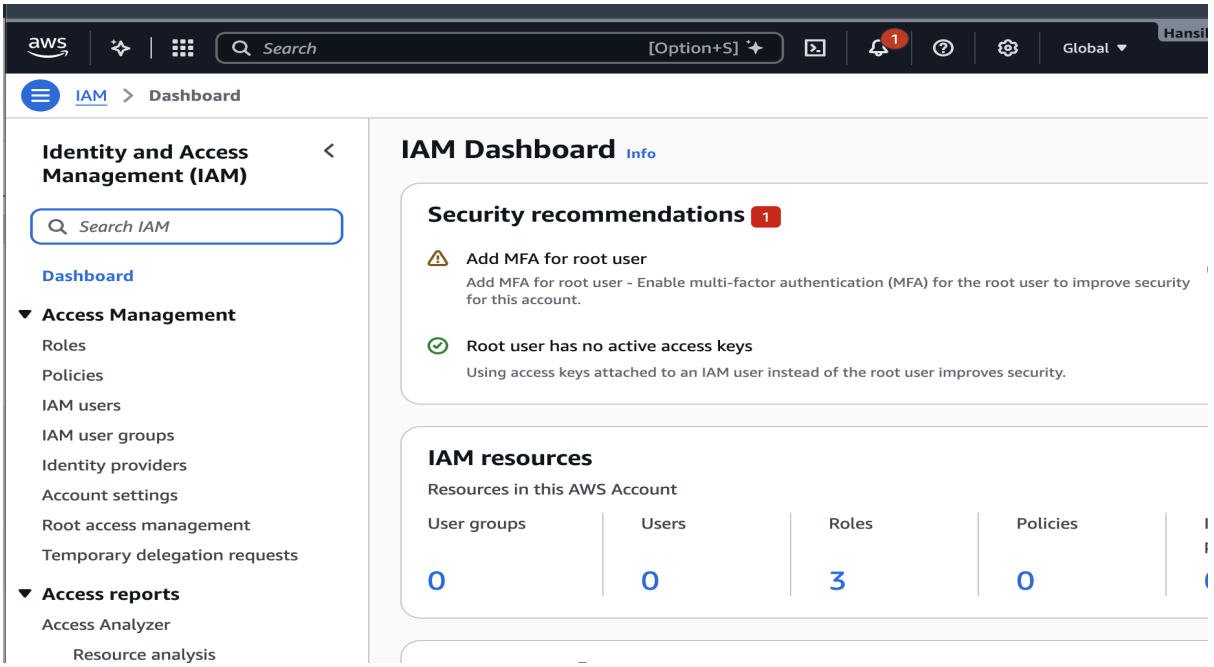
Step 1: Sign in as the root user

1. Open a web browser and go to the AWS Management Console.
2. Sign in using the root user credentials.



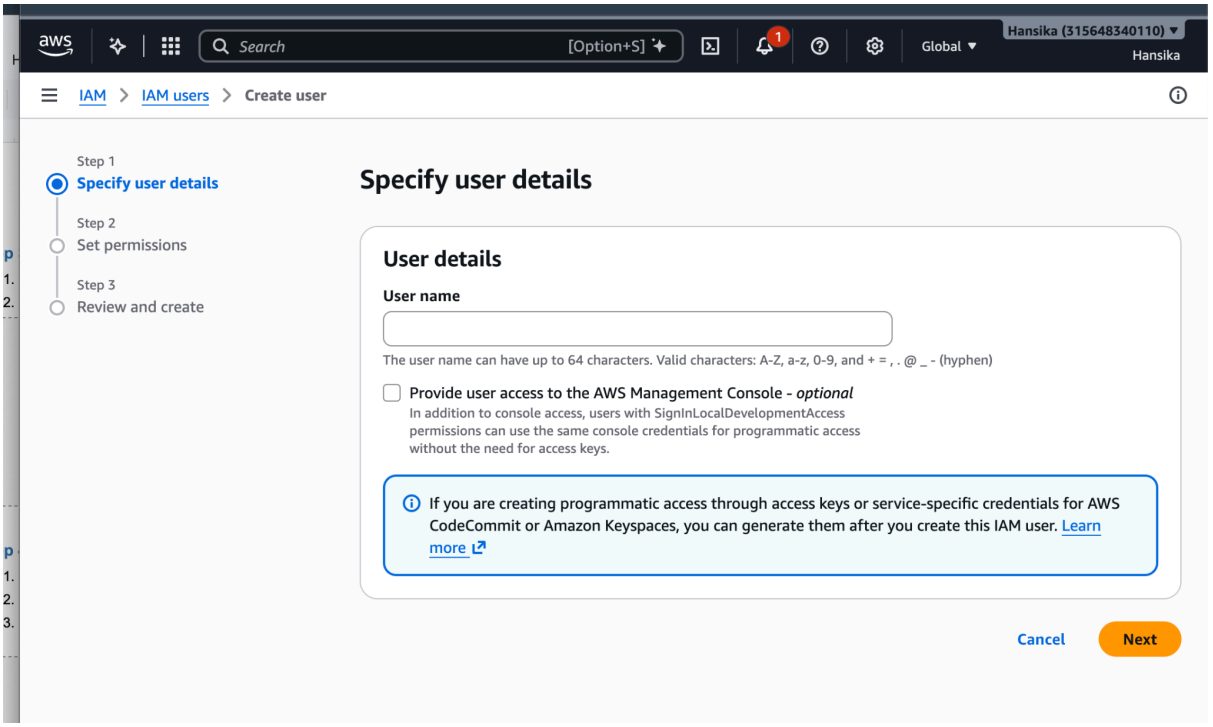
Step 2: Open the IAM service

1. In the search bar of the console, type IAM.
2. Select Identity and Access Management (IAM).



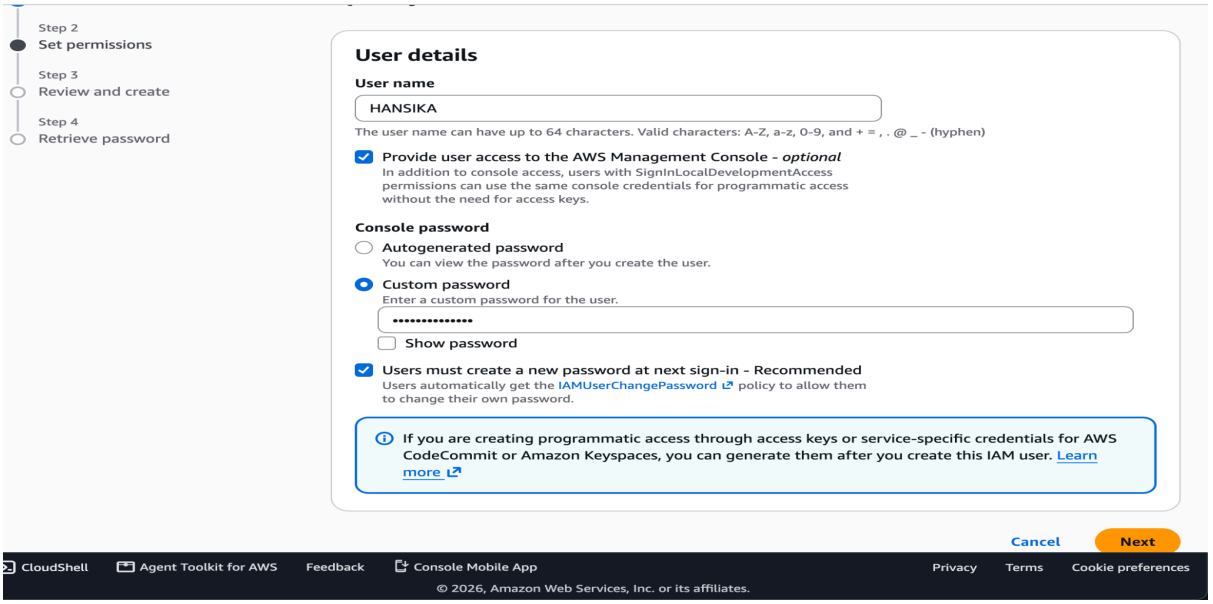
Step 3: Start creating the user

1. From the left navigation pane, select Users.
2. Click Create user.



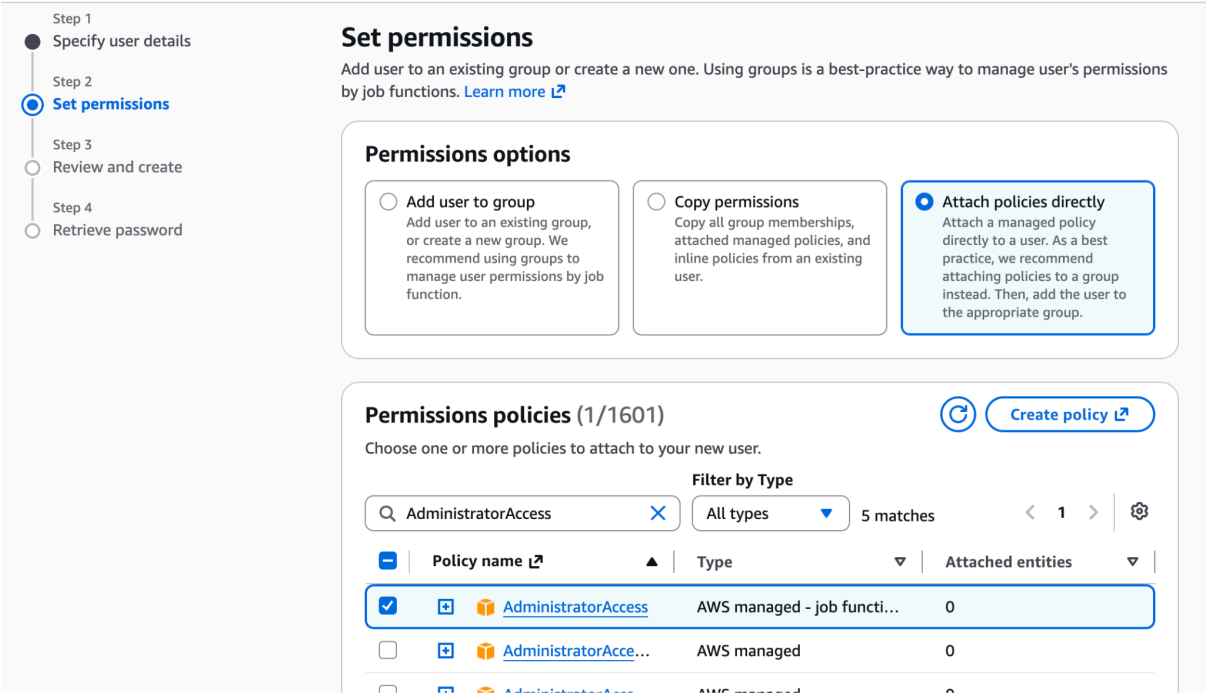
Step 4: Enter the user details

1. Enter a username (e.g., StudentUser01).
2. Select Provide user access to the AWS Management Console.
3. Set a custom password or choose an autogenerated password, optionally require a new password at next sign-in, and click Next.



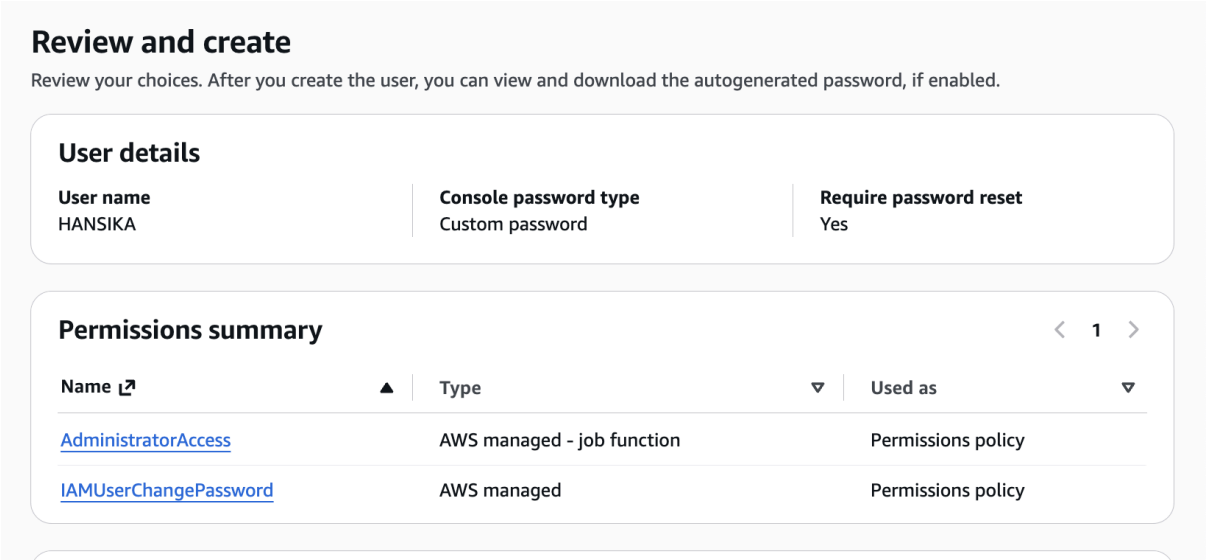
Step 5: Set the permissions

1. Under Set permissions, select Attach policies directly.
2. Select AdministratorAccess (AWS managed - job function) and click Next.



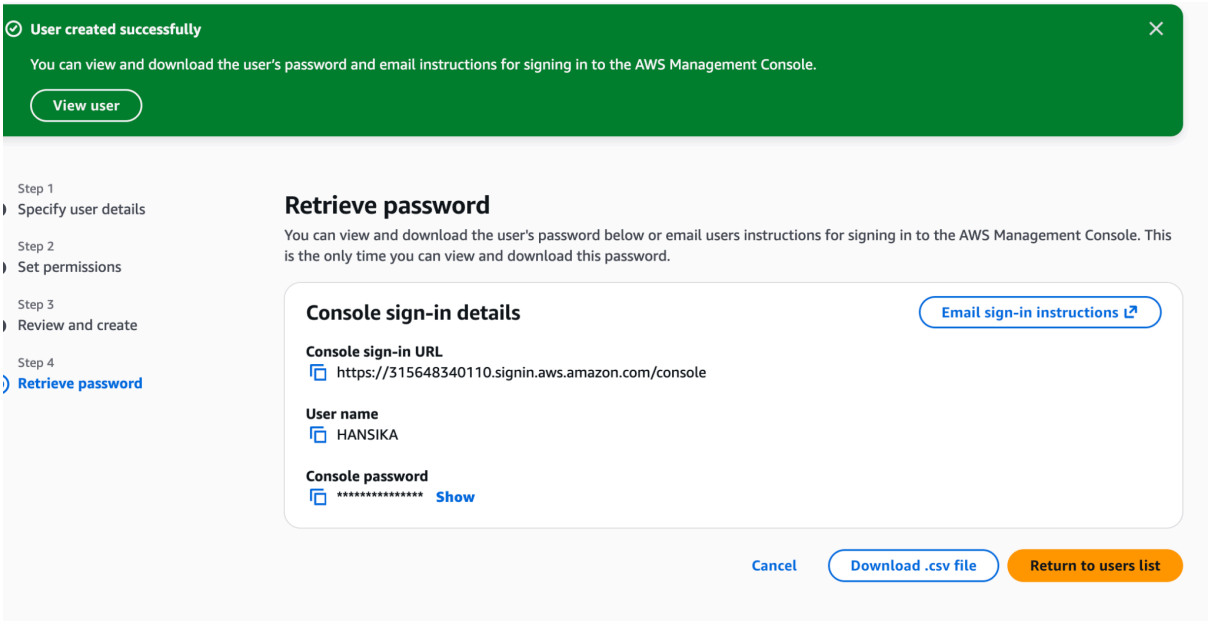
Step 6: Review and create the user

1. Review the username and permission details (tags are optional).
2. Click Create user.



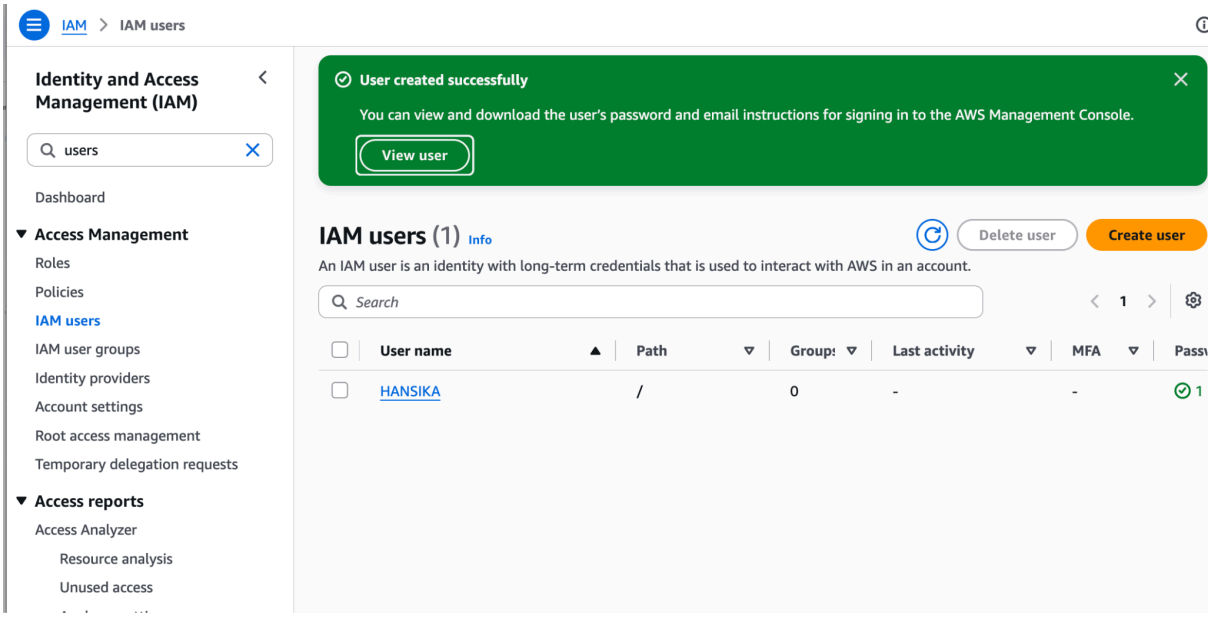
Step 7: Save the sign-in details

1. Note down the IAM user sign-in details.
2. Click Download .csv file to save the credentials securely.



Step 8: View the created user and sign out

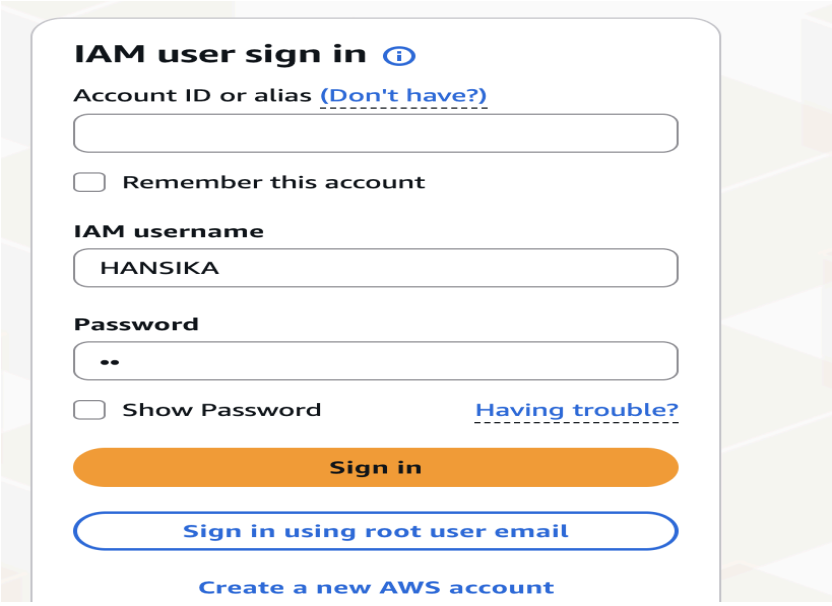
1. Click Return to users list and check that the new IAM user is listed.
2. Sign out of the root account.



Part B: Signing in as the IAM user

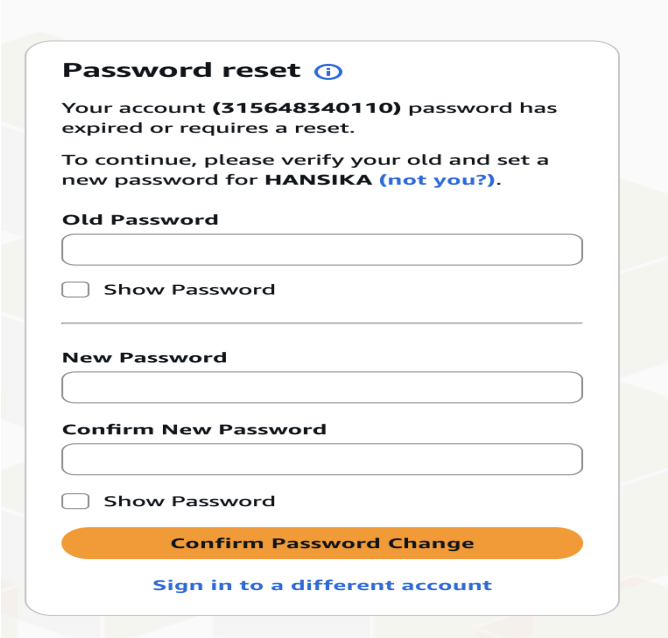
Step 9: Open the IAM user sign-in page

1. Open the AWS sign-in page and select IAM user (or open the Console sign-in URL saved in the .csv file).
2. Enter the account ID or alias (filled automatically when the .csv URL is used), then the IAM username and password.



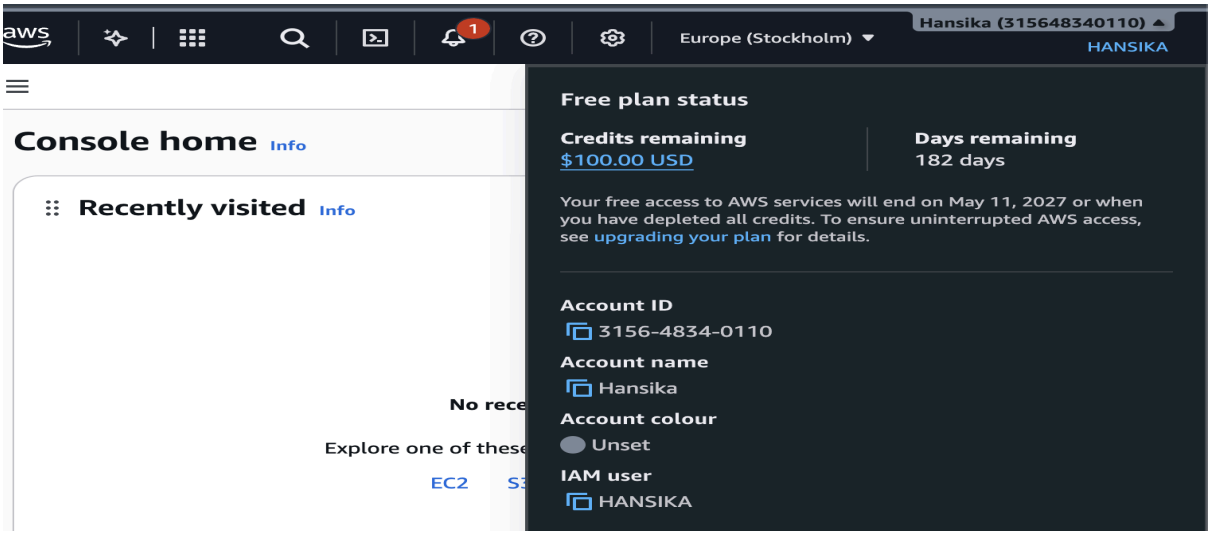
Step 10: Change the password (only if prompted)

1. If AWS asks, change the temporary password and set a new password.



Step 11: Verify access to the console

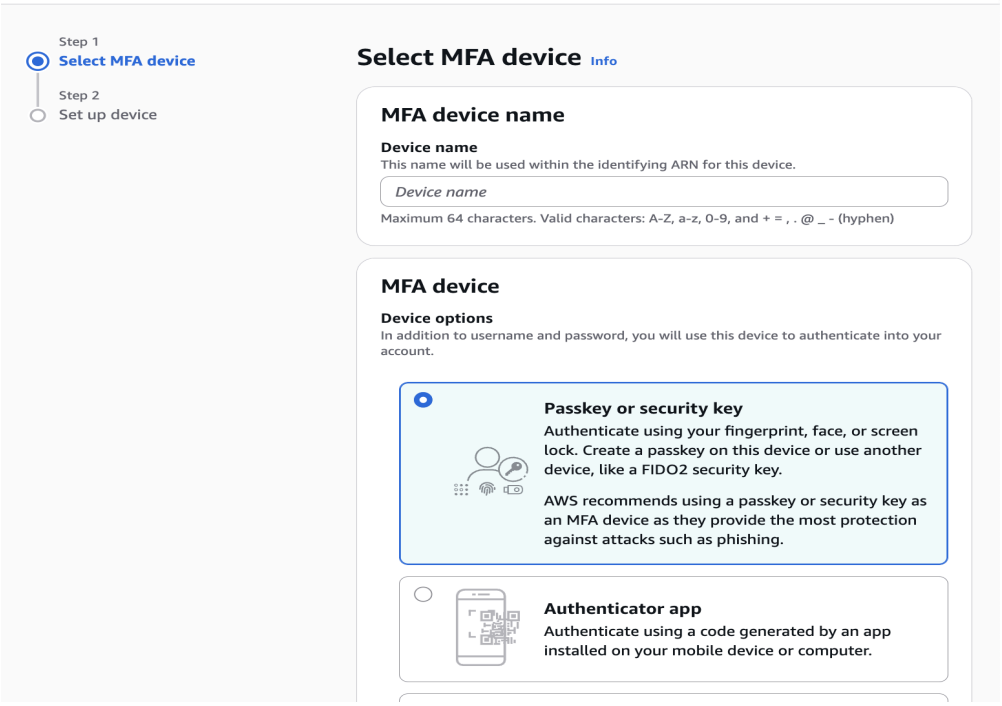
1. Verify that the AWS Management Console opens successfully and the IAM username is shown at the top-right.



Part C: Enabling Multi-Factor Authentication (MFA)

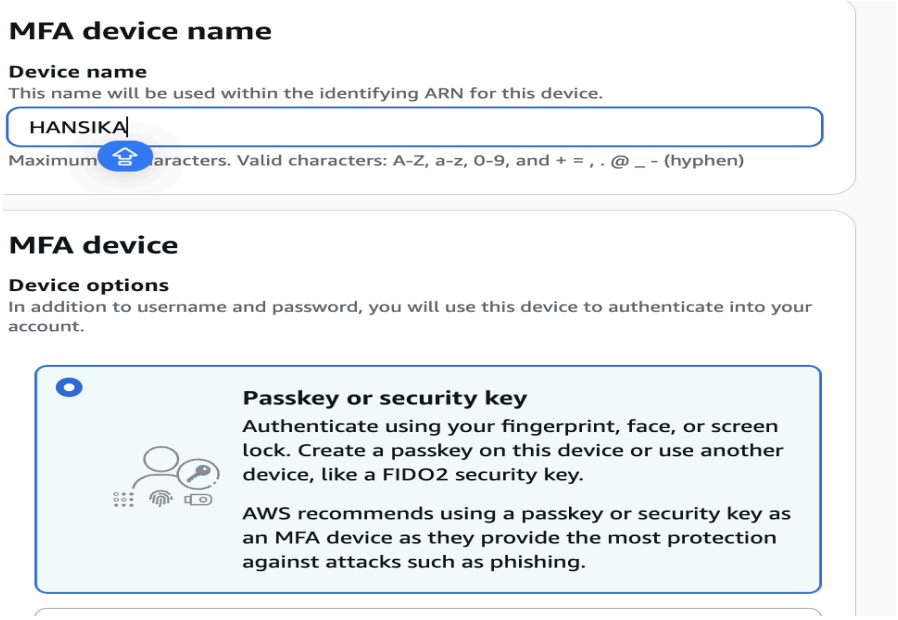
Step 12: Open the security credentials

1. In IAM, select Users and click the newly created IAM user.
2. Open the Security credentials tab and locate Multi-factor authentication (MFA).
3. Click Assign MFA device.



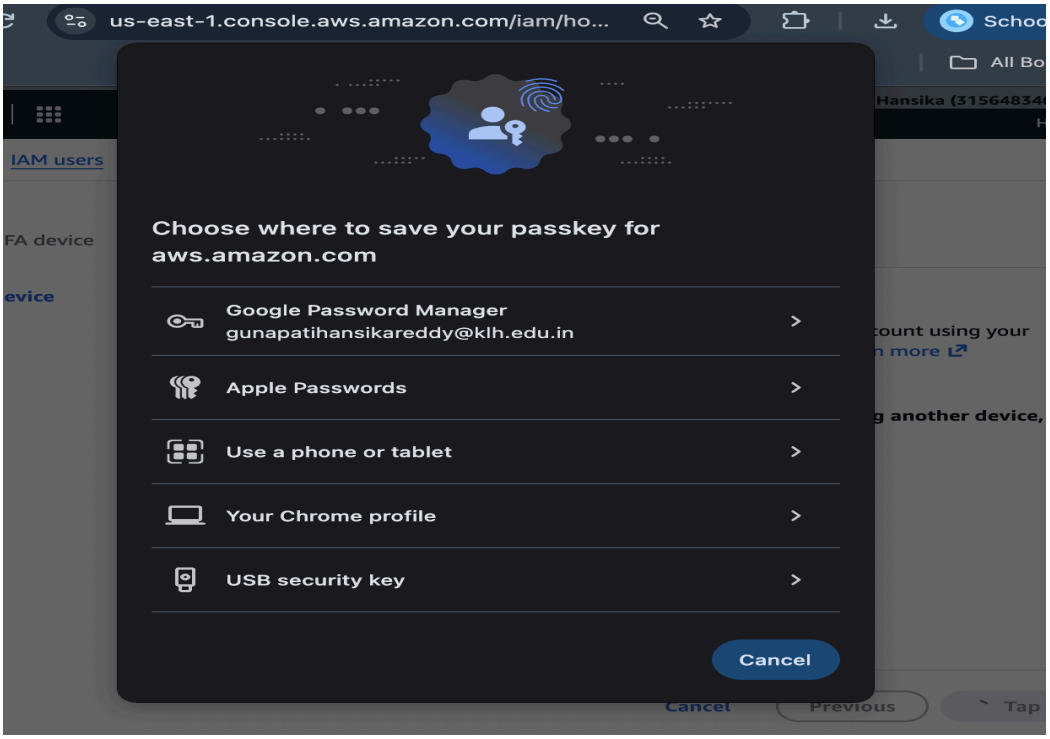
Step 13: Choose the MFA device type

1. Enter a device name (e.g., StudentMFA01).
2. Select the MFA device type: Passkey or Security key, or Authenticator app.

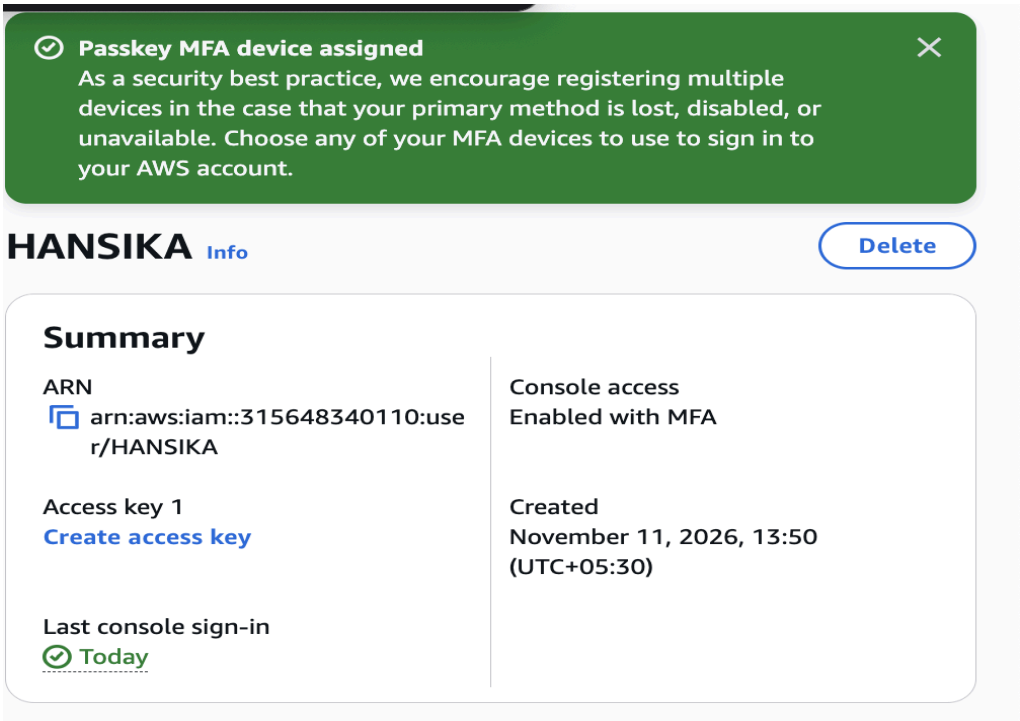


Step 14 (Option A): Register a Passkey or Security key

1. Select Passkey or Security key and click Next.
2. Follow the browser instructions to create the passkey (e.g., save it in Google Password Manager) and verify with biometrics, device PIN or security key.

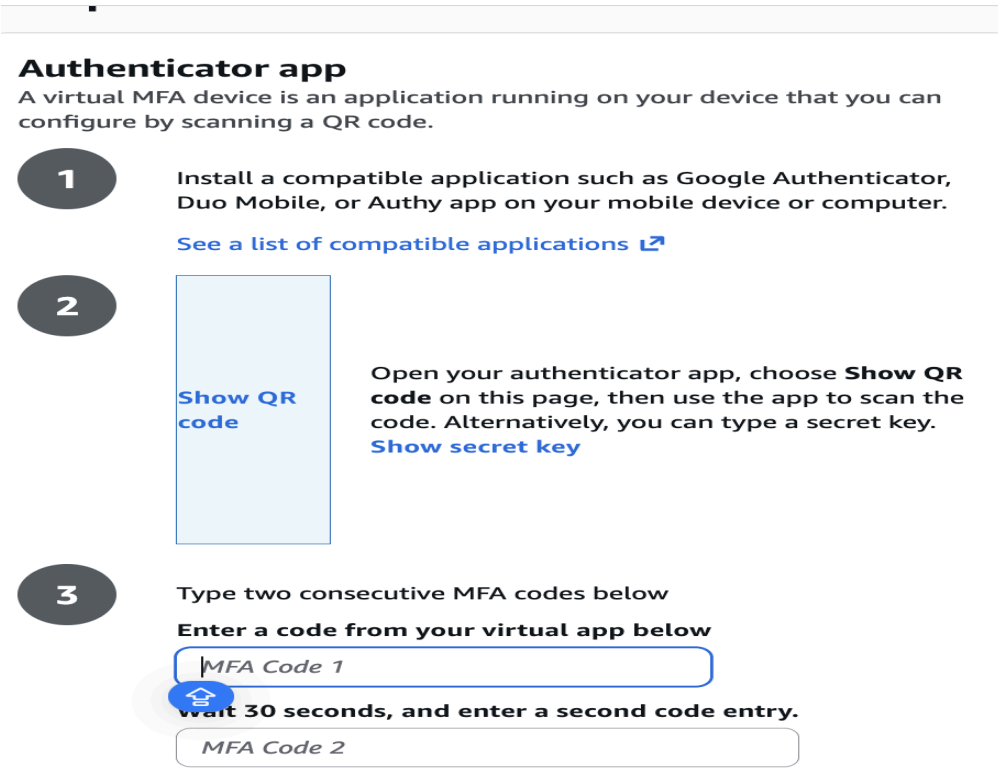


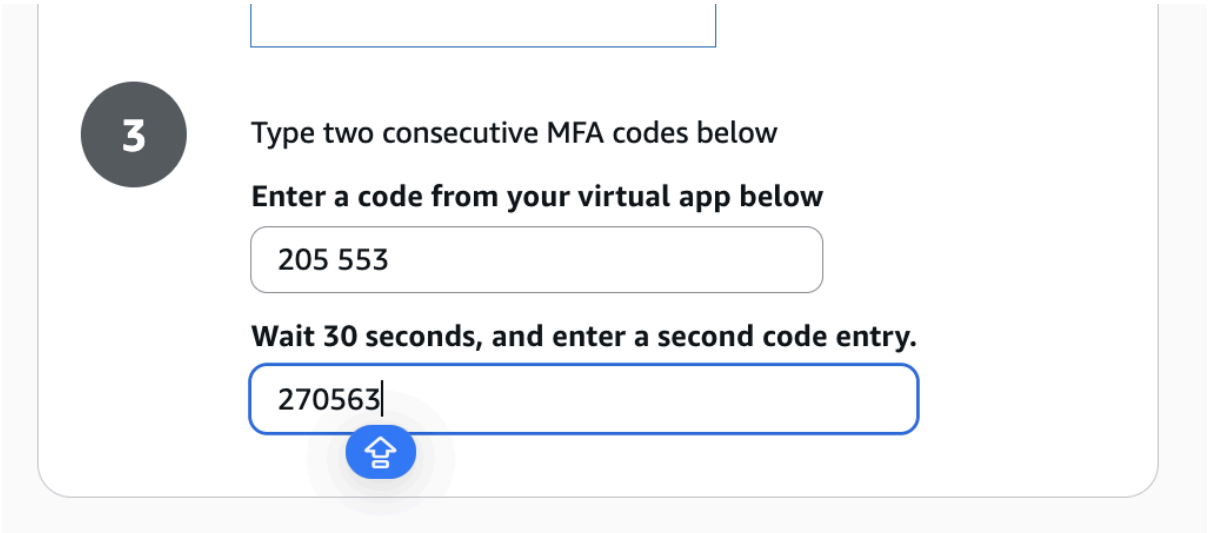
3. Click Continue to complete the registration.



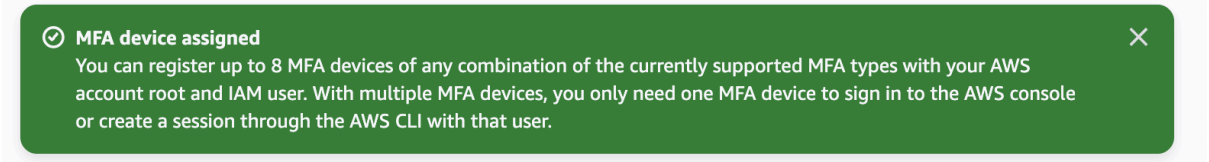
Step 14 (Option B): Register an Authenticator app

1. Select Authenticator app and click Next.
2. Open Google Authenticator or Microsoft Authenticator, choose add new account and scan the QR code shown in the AWS console.



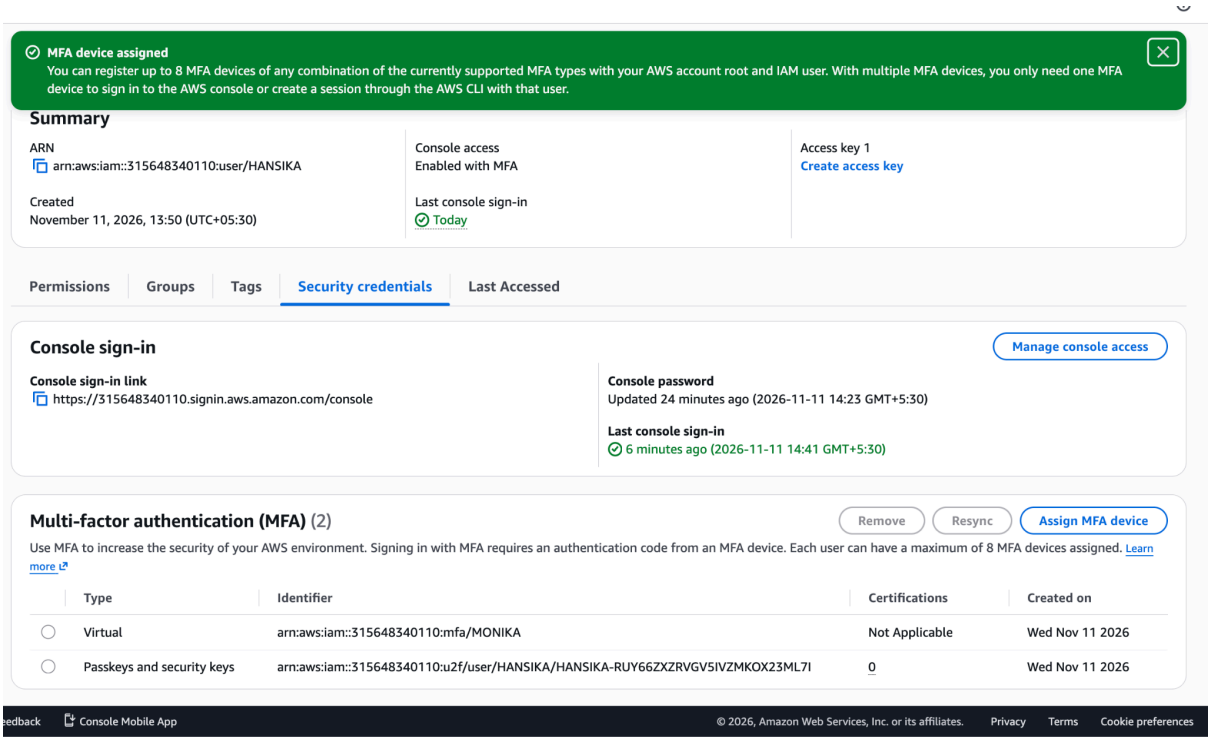


Enter the first six-digit code, wait for the next code, enter the second code and click Add MFA.



Step 15: Verify the MFA assignment

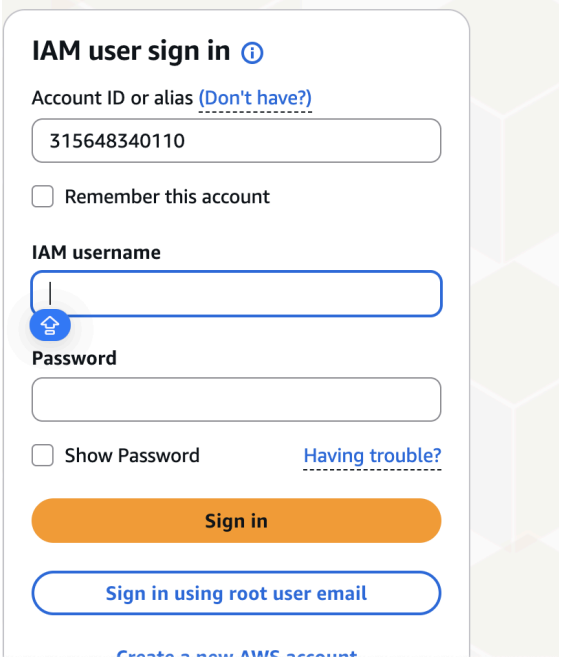
1. Verify that the selected MFA device is displayed as assigned to the IAM user.

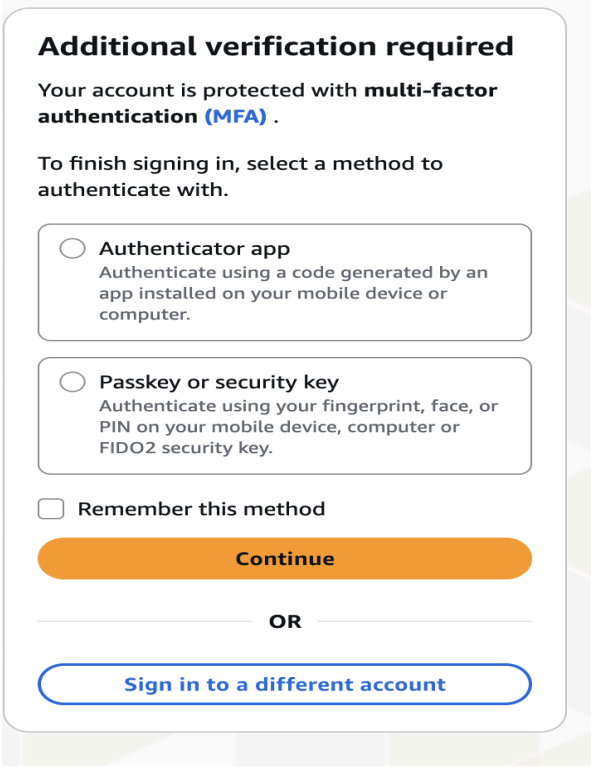


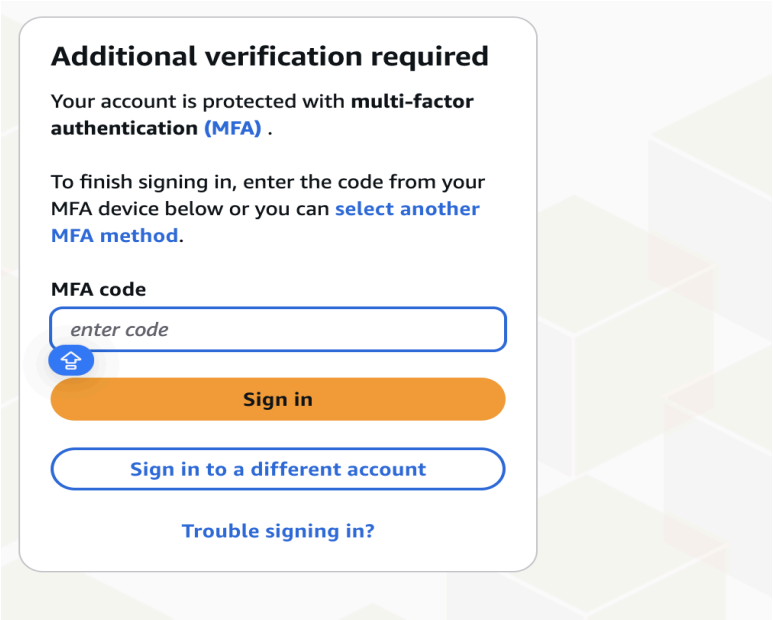
Part D: Testing MFA authentication

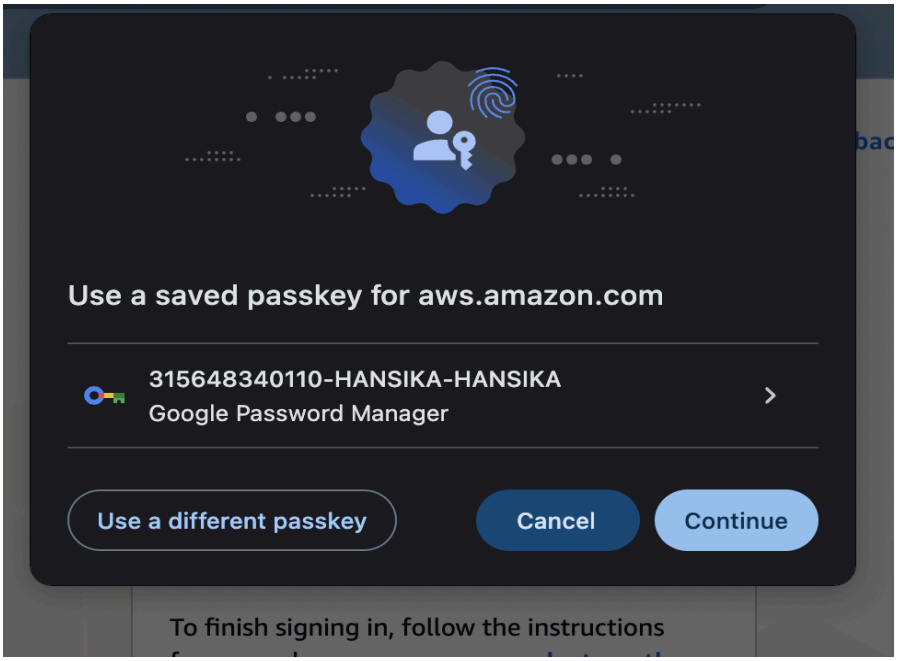
Step 16: Sign in again and complete MFA

1. Sign out of the AWS Management Console and open the AWS sign-in page.
2. Select IAM user, enter the account ID or alias, then the IAM username and password.
3. If a passkey was used, select the passkey and verify with the registered device, biometrics or PIN.
4. If an authenticator app was used, enter the current six-digit code shown in the app.



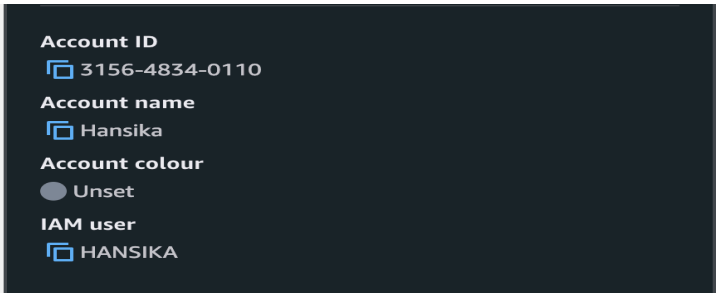






Step 17: Verify the login

1. Verify that the login to the AWS Management Console is successful after the MFA verification.



5. Observation

Parameter	Observation
IAM user created	StudentUser01 (with console access)
Permission assigned	AdministratorAccess (AWS managed - job function)
Sign-in as IAM user	Successful using the account ID/alias, username and password
MFA device type	Passkey / Security key or Authenticator app (as used)
MFA status	The MFA device was listed as assigned under Security credentials
Sign-in after MFA	AWS asked for the additional MFA verification; login was successful after completing it

6. Result

- An IAM user was successfully created and given the required permissions.
- Multi-Factor Authentication was successfully enabled for the IAM user.
- The IAM user could access the AWS Management Console only after completing the MFA verification.

7. Conclusion

Multi-Factor Authentication adds a second layer of protection on top of the username and password. Even if the password is exposed, the account cannot be accessed without the second factor, so MFA should be enabled for every user, especially those with administrator permissions.